



Flower Classification Using Transfer Learning Models

Course Project Report

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1. Introduction

Image classification is one of the most important applications of deep learning and computer vision. It enables computers to automatically recognize and categorize objects within images, making it useful in many fields such as healthcare, agriculture, security, and environmental monitoring.

Flower classification is a challenging image classification task because many flower species share similar visual characteristics, including color, shape, and texture. Accurate flower recognition requires powerful feature extraction techniques capable of capturing subtle differences between classes.

In recent years, transfer learning has become a highly effective approach for image classification problems. Instead of training deep neural networks from scratch, transfer learning utilizes pre-trained models that have already learned useful visual features from large-scale datasets such as ImageNet. This approach reduces training time and improves classification performance, especially when working with limited datasets.

In this project, the Oxford Flowers 102 dataset was used to develop and evaluate flower classification models. Two popular transfer learning architectures, MobileNetV2 and ResNet50V2, were selected and compared. Both models were trained using transfer learning techniques and evaluated using multiple performance metrics, including Accuracy, Precision, Recall, and F1-Score. In addition, confusion matrices and error analysis were performed to better understand model behavior and classification errors.

The objective of this project is to identify the most effective transfer learning model for flower classification and analyze its performance on the Oxford Flowers 102 dataset.

2. Dataset Description

The Oxford Flowers 102 dataset was used in this project as the primary dataset for flower image classification. This dataset is widely used in computer vision research and serves as a benchmark for evaluating image classification algorithms. It contains images collected from the internet representing a diverse set of flower species with significant variations in color, shape, texture, scale, background, and lighting conditions.

The dataset consists of 102 flower categories, where each category represents a distinct flower species. The images present several classification challenges because many flower classes exhibit similar visual characteristics, making accurate recognition a non-trivial task. Additionally, images were captured under different environmental conditions and from various viewpoints, increasing the complexity of the classification problem.

For this project, the dataset was loaded directly using TensorFlow Datasets (TFDS), which provides predefined training, validation, and testing splits. The dataset distribution used during the experiments is summarized in Table 1.

Dataset Split	Number of Images
Training Set	1,020
Validation Set	1,020
Test Set	6,149
Total	8,189

Table 1. Dataset Distribution

The training set was used to train the classification models, while the validation set was utilized to monitor model performance during training and support hyperparameter tuning. The test set was reserved exclusively for final evaluation to ensure an unbiased assessment of model performance.

Figure 1 illustrates a sample of images from the Oxford Flowers 102 dataset. As shown, the dataset contains flowers with diverse visual appearances, including variations in petal structure, color distribution, image background, and object scale. Such diversity makes the dataset suitable for evaluating the effectiveness of transfer learning models in complex image classification tasks.

The Oxford Flowers 102 dataset was selected because it provides a realistic and challenging classification problem while maintaining sufficient diversity to compare the performance of different deep learning architectures. Its popularity in academic research also enables meaningful benchmarking and comparison with previously reported results.



Figure 1. Sample images from the Oxford Flowers 102 dataset.

3. Data Preprocessing and Augmentation

Before training the deep learning models, several preprocessing and data augmentation techniques were applied to improve model performance and enhance generalization. These steps are essential in image classification tasks because deep neural networks require standardized inputs and sufficient data variability to learn robust visual features.

3.1 Image Preprocessing

The original images in the Oxford Flowers 102 dataset have different resolutions and dimensions. Since both MobileNetV2 and ResNet50V2 require a fixed input size, all images were resized to **224 × 224 pixels** before being fed into the models.

In addition, pixel values were normalized by converting image data from integer values in the range [0, 255] to floating-point values in the range [0, 1]. Normalization improves numerical stability during training and helps accelerate model convergence.

The preprocessing pipeline included the following operations:

- Image resizing to 224 × 224 pixels.
- Conversion of image data to floating-point format.
- Pixel value normalization to the range [0, 1].

These operations ensured that all images had a consistent representation suitable for transfer learning models.

3.2 Data Augmentation

Although the Oxford Flowers 102 dataset contains a large number of images, deep learning models can still suffer from overfitting when trained on limited variations of the training data. To address this issue, data augmentation techniques were applied during training.

Data augmentation artificially increases dataset diversity by generating modified versions of existing images while preserving their semantic content. This enables the model to become more robust to variations that may occur in real-world scenarios.

The following augmentation techniques were used:

Horizontal Flipping

Random horizontal flipping was applied to create mirrored versions of flower images. This helps the model learn orientation-invariant features.

Random Rotation

Images were randomly rotated within a small range. This allows the model to recognize flowers from different viewing angles.

Random Zoom

Random zoom transformations were applied to simulate variations in object scale and camera distance.

Random Contrast Adjustment

Random contrast modifications were introduced to improve robustness against lighting and image quality variations.

By applying these augmentation techniques, the training dataset became more diverse, reducing overfitting and improving the models' ability to generalize to unseen test images.

3.3 Dataset Pipeline Optimization

To improve training efficiency, TensorFlow data pipeline optimizations were used. The dataset was shuffled before training to ensure random sampling of images, batched into groups of 32 images, and prefetched using TensorFlow's AUTOTUNE functionality.

The final data pipeline consisted of:

- Shuffling the training dataset.
- Batching images with a batch size of 32.
- Prefetching data to optimize GPU utilization.

These optimizations reduced input bottlenecks and improved overall training performance.

4. Methodology

Transfer learning was adopted in this project to leverage the knowledge learned by deep convolutional neural networks that were previously trained on the large-scale ImageNet dataset. Instead of training a model from scratch, pre-trained models can extract high-level visual features efficiently, reducing training time and improving classification performance.

Two state-of-the-art transfer learning architectures were selected and evaluated:

1. MobileNetV2
2. ResNet50V2

Both models were initialized using pre-trained ImageNet weights and adapted to perform flower classification on the Oxford Flowers 102 dataset. The original classification layers of the pre-trained networks were removed and replaced with a new custom classification head designed specifically for the 102 flower categories.

The methodology consisted of two main stages:

- Frozen Transfer Learning
- Fine-Tuning

During the frozen transfer learning stage, all pre-trained layers were kept fixed while only the newly added classification layers were trained. In the fine-tuning stage, selected layers of the pre-trained network were unfrozen and further trained using a small learning rate to improve task-specific performance.

Dataset → Preprocessing → Data Augmentation → Transfer Learning Model → Fine-Tuning → Evaluation

4.1 MobileNetV2 Architecture

MobileNetV2 is a lightweight convolutional neural network architecture specifically designed for efficient image classification tasks. It utilizes depthwise separable convolutions and inverted residual blocks, significantly reducing computational complexity while maintaining strong classification performance.

In this project, MobileNetV2 pre-trained on the ImageNet dataset was used as the base feature extractor. The original classification layer was removed by setting the parameter `include_top=False`, allowing the model to generate high-level visual representations suitable for the flower classification task.

To adapt the model to the Oxford Flowers 102 dataset, a custom classification head was added consisting of:

- Global Average Pooling layer.
- Dropout layer with a rate of 0.3 to reduce overfitting.
- Fully connected Dense layer with 102 output neurons.
- Softmax activation function for multi-class classification.

Initially, all MobileNetV2 layers were frozen and only the custom classification head was trained. Subsequently, fine-tuning was performed by unfreezing a subset of the deeper layers and retraining the model using a very small learning rate to preserve previously learned visual features.

MobileNetV2 was selected because it offers an effective balance between computational efficiency and classification accuracy, making it a popular choice for transfer learning applications.

4.2 ResNet50V2 Architecture

ResNet50V2 is a deep residual convolutional neural network consisting of 50 layers. The architecture introduces residual learning through shortcut connections, which help mitigate the vanishing gradient problem and enable the successful training of deeper neural networks.

For this project, the ImageNet pre-trained ResNet50V2 model was employed as the second transfer learning architecture. Similar to MobileNetV2, the original classification head was removed and replaced with a custom classification layer suitable for the 102 flower categories.

The customized architecture included:

- Global Average Pooling layer.
- Dropout layer with a rate of 0.3.
- Dense output layer containing 102 neurons.
- Softmax activation function.

The training process was performed in two phases. First, the pre-trained backbone network was frozen and only the classification head was trained. After obtaining a stable baseline model, fine-tuning was applied by unfreezing the last layers of ResNet50V2 and retraining them using a low learning rate.

ResNet50V2 was selected because of its strong feature extraction capability and proven effectiveness in image classification tasks. Due to its deeper architecture and residual learning mechanism, it is capable of learning more complex visual representations compared to lightweight models.

5. Training Configuration

To ensure a fair comparison between MobileNetV2 and ResNet50V2, both models were trained using similar training settings and optimization strategies. The selected hyperparameters were chosen to balance training efficiency and model performance while reducing the risk of overfitting.

5.1 Optimizer

The Adam optimizer was used during all training experiments. Adam is one of the most widely used optimization algorithms in deep learning because it combines the advantages of momentum and adaptive learning rates, resulting in faster convergence and stable training behavior.

5.2 Loss Function

Since the task involves multi-class classification with 102 flower categories, the Sparse Categorical Cross-Entropy loss function was employed. This loss function is suitable when class labels are represented as integer values rather than one-hot encoded vectors.

5.3 Batch Size

A batch size of 32 images was used throughout training. This value provides a good balance between computational efficiency and stable gradient updates.

5.4 Learning Rate

During the initial transfer learning phase, a learning rate of 0.001 was used for training the newly added classification layers.

For the fine-tuning phase, a significantly smaller learning rate of 0.00001 was employed. Using a lower learning rate helps preserve the valuable features learned from ImageNet while allowing the model to adapt to the flower classification task.

5.5 Number of Epochs

Both MobileNetV2 and ResNet50V2 were initially trained for up to 10 epochs during the frozen transfer learning stage.

Additional fine-tuning epochs were performed after unfreezing selected layers of the pre-trained networks. This two-stage training strategy allowed the models to gradually adapt to the target dataset while minimizing overfitting.

5.6 Regularization Techniques

Several regularization methods were applied to improve generalization performance:

- Data augmentation techniques.
- Dropout layer with a dropout rate of 0.3.
- Early stopping mechanism.
- Learning rate reduction during training.

These techniques helped reduce overfitting and improved the models' ability to generalize to unseen test images.

5.7 Training Callbacks

Two TensorFlow callbacks were utilized during training:

Early Stopping

Early Stopping monitored validation performance and automatically terminated training when no further improvement was observed. The best-performing model weights were restored to prevent performance degradation caused by overfitting.

ReduceLROnPlateau

The ReduceLROnPlateau callback automatically decreased the learning rate whenever validation performance stagnated. This enabled the models to perform more refined parameter updates during later stages of training.

Table 2 summarizes the main training hyperparameters used throughout the experiments.

Parameter	Value
Optimizer	Adam
Initial Learning Rate	0.001
Fine-Tuning Learning Rate	0.00001
Batch Size	32
Frozen Training Epochs	10
Fine-Tuning Epochs	5
Loss Function	Sparse Categorical Cross-Entropy
Dropout Rate	0.3

Table 2. Training Hyperparameters

6. Evaluation Metrics

To objectively evaluate the performance of the developed flower classification models, several commonly used classification metrics were employed. Relying on a single metric may not provide a complete understanding of model behavior; therefore, multiple evaluation measures were used to assess different aspects of classification performance.

The evaluation was conducted using Accuracy, Precision, Recall, and F1-Score. In addition, confusion matrices were generated to visualize the classification results and identify common sources of misclassification.

6.1 Accuracy

Accuracy measures the proportion of correctly classified samples among all test samples. It provides an overall indication of model performance and is one of the most widely used evaluation metrics in classification tasks.

The accuracy is calculated as:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

A higher accuracy value indicates better overall classification performance.

6.2 Precision

Precision measures the proportion of correctly predicted positive samples among all samples predicted as positive. It reflects the reliability of the model's positive predictions.

The precision is calculated as:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

High precision indicates that the model makes fewer false positive predictions.

6.3 Recall

Recall measures the ability of the model to correctly identify all relevant samples belonging to a class.

The recall is calculated as:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

A high recall value indicates that the model successfully detects most of the actual samples belonging to a class.

6.4 F1-Score

The F1-Score Combines Precision and Recall into a single metric by computing their harmonic mean.

The F1-Score is calculated as:

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The F1-Score is particularly useful when evaluating multi-class classification problems because it balances the trade-off between Precision and Recall.

Since the Oxford Flowers 102 dataset contains a large number of flower categories, the Macro F1-Score was used. Macro averaging computes the F1-Score independently for each class and then calculates the average across all classes, giving equal importance to every flower category.

6.5 Confusion Matrix

A confusion matrix was used to provide a detailed visualization of classification performance across all flower categories.

The rows of the confusion matrix represent the actual classes, while the columns represent the predicted classes. Correct classifications appear along the main diagonal, whereas off-diagonal elements indicate classification errors.

Confusion matrices are valuable for identifying patterns of misclassification and understanding which flower categories are frequently confused by the model.

6.6 Error Analysis

In addition to numerical metrics, an error analysis procedure was performed by examining misclassified test images. This analysis helps identify challenging flower categories and provides insights into the limitations of the classification models.

Studying incorrectly classified samples allows a deeper understanding of model weaknesses and highlights potential areas for future improvement.

7. Experimental Results

This section presents the performance results obtained from the two transfer learning models used in this project: MobileNetV2 and ResNet50V2. The models were evaluated on the Oxford Flowers 102 test dataset using Accuracy, Precision, Recall, and Macro F1-Score.

7.1 MobileNetV2 Results

MobileNetV2 was first trained using transfer learning with frozen pre-trained layers and subsequently fine-tuned to improve task-specific performance. The model demonstrated strong classification capability while maintaining a relatively lightweight architecture.

Table 3 summarizes the final evaluation results obtained using MobileNetV2.

Metric	Value
Accuracy	73.13%
Precision	73.64%
Recall	74.22%
Macro F1-Score	71.94%

Table 3. MobileNetV2 Performance Metrics

The results indicate that MobileNetV2 successfully learned discriminative features for flower classification and achieved good overall performance. The model showed balanced Precision and Recall values, indicating stable classification behavior across multiple flower categories.

Figure 2 presents the confusion matrix generated for MobileNetV2.

Figure 2. MobileNetV2 Confusion Matrix

The confusion matrix shows a strong diagonal pattern, indicating that the majority of flower images were correctly classified. However, several off-diagonal entries are visible, suggesting that certain flower categories with similar visual characteristics were occasionally misclassified.

7.2 ResNet50V2 Results

ResNet50V2 was evaluated using the same training and testing procedures. Due to its deeper architecture and residual learning mechanism, the model was expected to capture more complex visual features.

After transfer learning and fine-tuning, ResNet50V2 achieved the results shown in Table 4.

Metric	Value
Accuracy	75.38%
Precision	72.34%
Recall	76.59%
Macro F1-Score	73.19%

Table 4. ResNet50V2 Performance Metrics

The results demonstrate that ResNet50V2 achieved superior classification performance compared with MobileNetV2. Although the Precision value was slightly lower, the model obtained higher Recall and Macro F1-Score values, indicating improved overall recognition capability across the 102 flower categories.

Figure 3 illustrates the confusion matrix generated for ResNet50V2.

Figure 3. ResNet50V2 Confusion Matrix

The confusion matrix exhibits a clearer diagonal structure compared to MobileNetV2, suggesting improved classification accuracy. Nevertheless, some confusion remained between visually similar flower species, which is expected given the complexity of the dataset.

7.3 Model Comparison

To identify the best-performing architecture, the results of both models were compared using the selected evaluation metrics.

Model	Accuracy	Precision	Recall	Macro F1
MobileNetV2	73.13%	73.64%	74.22%	71.94%
ResNet50V2	75.38%	72.34%	76.59%	73.19%

Table 5. Performance Comparison Between Models

The comparison reveals that ResNet50V2 achieved the highest Accuracy and Macro F1-Score among the evaluated models. Although MobileNetV2 provided competitive performance with lower computational complexity, ResNet50V2 demonstrated stronger feature extraction capabilities and better overall classification performance.

Based on these results, Fine-Tuned ResNet50V2 was selected as the best-performing model for the flower classification task.

8. Error Analysis

Although the proposed transfer learning models achieved satisfactory classification performance, some classification errors were still observed during testing. To better understand the limitations of the models, an error analysis was performed by examining a number of misclassified flower images generated by the best-performing model, Fine-Tuned ResNet50V2.

Figure 4 presents several examples of incorrectly classified images from the test dataset.

Figure 4. Examples of Misclassified Flower Images

The analysis of misclassified samples revealed several common factors that contributed to prediction errors.

8.1 Visual Similarity Between Flower Categories

One of the primary sources of misclassification was the high visual similarity between certain flower species. Many flower categories share similar colors, petal structures, and overall shapes, making it difficult for the model to distinguish between them.

In such cases, even minor differences in flower appearance can lead to incorrect predictions.

8.2 Complex Backgrounds

Some images contained cluttered or highly textured backgrounds that distracted the model from focusing on the flower itself. Background elements such as leaves, grass, soil, and surrounding vegetation occasionally introduced additional visual patterns that affected classification performance.

8.3 Variations in Illumination

The dataset includes images captured under different lighting conditions. Variations in brightness, shadows, and contrast sometimes altered the appearance of flowers and increased classification difficulty.

These lighting differences may cause flowers belonging to the same class to appear visually different from one another.

8.4 Scale and Viewpoint Variations

Flowers were photographed from different distances and viewing angles. In some images, the flower occupied only a small portion of the frame, while in others it appeared as a close-up image.

Such variations can reduce the visibility of important distinguishing features and increase the likelihood of misclassification.

8.5 Model Limitations

Although ResNet50V2 achieved the best overall performance, it is still limited by the information available in the training dataset. Some flower categories contain only subtle visual differences that may require larger datasets, more advanced architectures, or additional fine-tuning strategies to achieve higher classification accuracy.

Overall, the error analysis demonstrates that most misclassifications occurred in challenging scenarios involving visually similar flower species, complex backgrounds, and variations in image acquisition conditions. Despite these challenges, the model maintained strong overall performance and successfully classified the majority of test images.

9. Conclusion

This project investigated the application of transfer learning techniques for flower image classification using the Oxford Flowers 102 dataset. Two pre-trained deep learning architectures, MobileNetV2 and ResNet50V2, were implemented, trained, fine-tuned, and evaluated using multiple performance metrics.

A complete classification pipeline was developed, including data preprocessing, data augmentation, transfer learning, fine-tuning, model evaluation, confusion matrix analysis, and error analysis. These steps enabled the models to learn discriminative visual features and effectively classify flower species across 102 categories.

Experimental results demonstrated that both models achieved good classification performance. MobileNetV2 provided competitive results while maintaining a lightweight architecture and lower computational complexity. However, ResNet50V2 achieved superior overall performance due to its deeper architecture and stronger feature extraction capability.

The best-performing model was Fine-Tuned ResNet50V2, which achieved:

- **Accuracy:** 75.38%
- **Precision:** 72.34%

- **Recall:** 76.59%
- **Macro F1-Score:** 73.19%

The confusion matrix and error analysis revealed that most classification errors occurred between visually similar flower species and in images containing challenging backgrounds, lighting variations, or different viewpoints.

Overall, the results confirm the effectiveness of transfer learning for image classification tasks and demonstrate that pre-trained deep learning models can successfully classify flower species with high accuracy. Among the evaluated architectures, Fine-Tuned ResNet50V2 was selected as the final model because it provided the best overall classification performance on the Oxford Flowers 102 dataset.

Future work may include experimenting with more advanced architectures, larger image resolutions, additional data augmentation strategies, and ensemble learning approaches to further improve classification accuracy.

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